JAYANTH N

Data Scientist | GenAI & Agentic AI Engineer | ML Practitioner

jayanthmails@gmail.com | +91-8310457656 | Bengaluru / Hyderabad (Open to Relocate) | linkedin.com/in/jayanth-n077057208/

CAREER OBJECTIVE

Results-driven Data Scientist with hands-on experience in Generative AI, Agentic AI workflows, and classical Machine Learning. Seeking to build and deploy intelligent, grounded AI solutions—RAG pipelines, LLM-powered agents, and ML classification models—that deliver measurable business outcomes. Adept in LangChain, LangGraph, Azure OpenAI, and MLflow with a strong focus on responsible AI, prompt engineering, structured outputs, and evaluation frameworks.

PROFESSIONAL EXPERIENCE

Data Scientist / AI Engineer | ANTS IT Solutions Pvt. Ltd. — Hyderabad *Mar 2025 – Present (1 Year)*

- Designed and delivered end-to-end GenAI solutions leveraging RAG architecture with grounded responses, citations, and fallback behavior on Azure OpenAI.
- Built agentic AI workflows using LangGraph with tool/function calling, multi-step planning, routing, and verification for complex business tasks.
- Engineered prompts for consistent JSON structured outputs and implemented policy and content guardrails to reduce hallucination risk.
- Implemented chunking strategies, metadata filters, and hybrid retrieval (Azure AI Search) to improve answer faithfulness and relevance scores.
- Developed classical ML classification models with robust validation, leakage prevention, threshold optimization, and SHAP-based explainability.
- Applied MLflow for experiment tracking, prompt versioning, and offline evaluation sets to benchmark quality, latency, and cost trade-offs.
- Created data quality checks and monitoring signals for model drift, missingness detection, and performance regression over time.
- Collaborated with stakeholders to define measurable success metrics, acceptance criteria, and documentation for audit readiness.

Data Science Intern | Edukron — Bengaluru *Oct 2024 – Mar 2025 (6 Months)*

- Conducted end-to-end Exploratory Data Analysis (EDA) on real-world business datasets to uncover patterns, correlations, and data quality issues.
- Engineered domain-relevant features using Python (Pandas, NumPy) to improve predictive model performance.
- Built, trained, and evaluated supervised ML models using scikit-learn; documented accuracy, precision, recall, and F1 metrics.
- Developed stakeholder-ready dashboards and visualizations using Matplotlib, Seaborn, and Power BI to communicate insights clearly.
- Maintained clean, reproducible data workflows in Jupyter Notebooks and performed regular data quality validation checks.

TECHNICAL SKILLS

Programming: Python, SQL, Git

Machine Learning: scikit-learn, XGBoost, LightGBM, Model Evaluation (AUC, F1, Precision, Recall), Imbalance Handling

Generative AI & LLMs: RAG, Embeddings, Hybrid Retrieval, Chunking Strategies, Prompt Engineering, Citations, Guardrails, Structured Outputs (JSON)

Agentic AI: LangGraph, LangChain, Tool/Function Calling, Planning & Routing, Memory Patterns, ReAct-style Orchestration

NLP / LLMs: Token Limits, Context Windows, Prompt Patterns, Azure OpenAI

Frameworks: LangChain, LlamaIndex, FastAPI / Flask, Pydantic

Vector DB / Search: Azure AI Search (Vector), FAISS, Chroma

Azure: Azure OpenAI, Azure ML, Blob Storage / ADLS, Key Vault, App Service

MLops & Evals: MLflow, Prompt Evaluation, LLM Evals, Offline Test Sets, Monitoring, Drift Detection

Data & Visualization: Pandas, NumPy, EDA, Feature Engineering, Matplotlib, Seaborn, Power BI

KEY PROJECTS

Project 1: GenAI Agentic Shopping & Product Discovery Assistant | Retail Industry

Role: GenAI / Agentic AI Engineer **Tools:** Python, Azure OpenAI, LangChain, LangGraph, Azure AI Search, Embeddings, Prompt Engineering, Key Vault

Developed an intelligent GenAI and Agentic AI assistant for an APAC retail client that answers complex product queries and recommends alternatives using grounded retrieval over catalog, policy, and FAQ documents. The system features multi-step agent planning, tool calling, citation generation, and safety guardrails for production-grade deployment.

Roles & Responsibilities:

- Architected a multi-agent system using LangGraph with tool/function calling for product lookup, policy verification, and availability checks.
- Implemented RAG ingestion pipeline including catalog/FAQ chunking strategies and metadata-based filtering for precision retrieval.
- Designed prompts for grounded responses with citations, refusal logic, and structured JSON outputs for seamless UI rendering.
- Improved retrieval accuracy using hybrid search, reranking, and query rewriting for ambiguous or complex user queries.
- Created offline evaluation sets (golden Q&A) measuring faithfulness and relevance to guide iterative prompt and retrieval improvements.
- Integrated observability logs tracking latency, cost, and error types; established rollout checks for safe production deployment.

Project 2: Retail Product Category Classification using Machine Learning

Tools: Python, Pandas, NumPy, Scikit-learn, NLP, TF-IDF Vectorization, Logistic Regression, Random Forest, Naive Bayes, Feature Engineering, Data Cleaning, Matplotlib, Seaborn, Jupyter Notebook

Developed a machine learning classification system to automatically categorize retail products into predefined categories (e.g., electronics, clothing, groceries, home essentials) based on product attributes such as name, description, price, and historical sales data. This system helps retail businesses streamline product catalog management, improve search accuracy, and enhance customer shopping experience.

Roles & Responsibilities:

- Collected and prepared retail product dataset with labels and attributes.
- Performed data cleaning and handled missing or inconsistent product data.
- Applied NLP techniques to process product titles and descriptions.
- Converted text data into numerical features using TF-IDF vectorization.
- Built classification models such as Logistic Regression, Naive Bayes, and Random Forest.
- Evaluated model performance using accuracy, precision, recall, and F1-score.
- Tuned hyperparameters to improve classification accuracy.
- Selected best model for final product category prediction.
- Visualized classification results and category distribution insights.

EDUCATION

B.E. — Computer Science & Engineering | City Engineering College — Bengaluru, Karnataka

2021 – 2025 **|** *CGPA: 7.08 / 10.0*

12th — CBSE Board | Sri Chaitanya PU College — Bengaluru

2019 – 2021 **|** *Score: 81%*

CBSE — Central Board of Secondary Education | BEL Vidyalaya CBSE — Bengaluru

2019 **|** *Score: 78%*